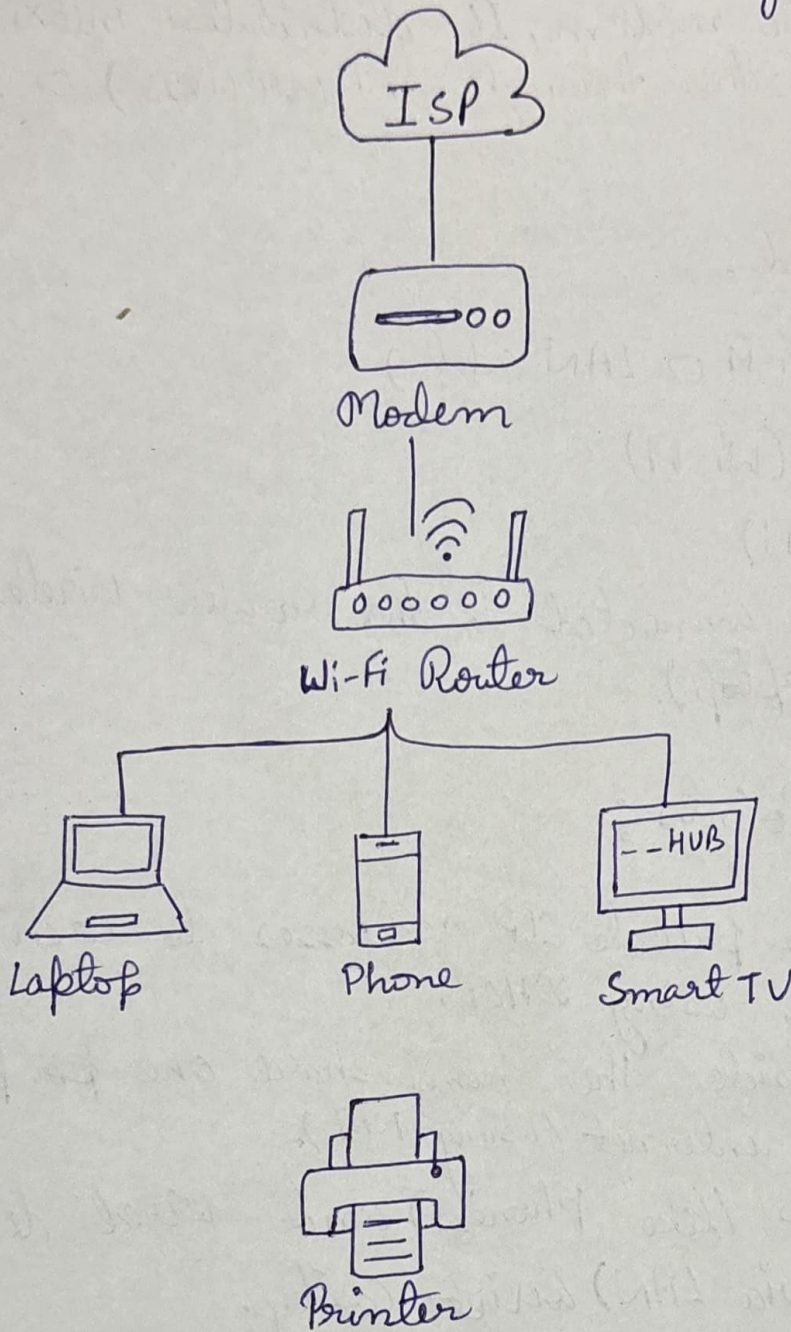


Assignment - 1

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• Explanation:-

→ ISP (Internet Service Provider):-

Provides internet access to your home. E.g:- Jio, Airtel, BSNL.

→ Modem:-

Converts the signal from the ISP into a digital signal your home devices can use. It connects directly to the ISP through a wired line (fiber optic/cable/DSL).

→ Wi-Fi Router:-

Connected to the modem, It distributes internet to multiple devices either through Wi-Fi (wireless) or LAN Cables (wired).

→ Devices Connected:-

- Laptop (via Wi-Fi or LAN cable)
- Mobile Phones (Wi-Fi)
- Smart TV (Wi-Fi)
- Printer (can be connected to the router wirelessly or wired to the laptop).

• Important Points:-

- Router assigns private IP addresses to each device inside the house using DHCP.
- All devices inside the home share one ~~for~~ public IP address on the internet (using NAT).
- Wireless devices (like phones) and wired devices (like a laptop via LAN) coexist easily.